9.16 Type Conversion

The process of converting one fundamental data type into another is called as "Type Conversion". C++ provides two types of conversions.

(1) Implicit type conversion

(2) Explicit type conversion.

(1) Implicit type conversion:

An Implicit type conversion is a conversion performed by the compiler automatically. So, implicit conversion is also called as **"Automatic conversion".**

This type of conversion is applied usually whenever different data types are intermixed in an expression. If the type of the operands differ, the compiler converts one of them to match with the other, using the rule that the "smaller" type is converted to the "wider" type, which is called as **"Type Promotion".**

For example:

```
#include <iostream>
using namespace std;
int main()
{
        int a=6;
        float b=3.14;
        cout << a+b;
}
```

In the above program, operand **a** is an int type and **b** is a float type. During the execution of the program, int is converted into a float, because a float is wider than int. Hence, the output of the above program will be: **9.14**

The following Table 9.11 shows you the conversion pattern.

RHO / LHO	**char**	**short**	**int**	**long**	**float**	**double**	**long double**
char	int	int	int	long	float	double	long double
short	int	int	int	long	float	double	long double
int	int	int	int	long	float	double	long double
long	long	long	long	long	float	double	long double
float	float	float	float	float	float	double	long double
double	double	double	double	double	double	double	long double
long double	long double	long double	long double	long double	long double	long double	long double

(RHO – Right Hand Operand; LHO – Left Hand Operand)

Table 9.11: Implicit conversion of mixed operands

11th Std Computer Science_EM Chapter 9.indd 147 06-01-2023 15:08:46